

From the Hopf bundle $S^1 \rightarrow S^3 \rightarrow S^2$ we have $\pi_4(S^2) \approx \pi_4(S^3) \approx \mathbb{Z}_2$, with $\pi_4(S^2)$ generated by the composition $\eta(\Sigma\eta)$ where η is the Hopf map $S^3 \rightarrow S^2$. From exactness of the latter part of the sequence we deduce that the map $\pi_4(S^4) \rightarrow \pi_3(S^2)$ is injective, and hence that the suspension $\pi_4(S^2) \rightarrow \pi_5(S^3)$ is surjective, so $\pi_5(S^3)$ is either $\mathbb{Z}_2$ or 0. From the general suspension theorem, the suspension $\pi_5(S^3) \rightarrow \pi_6(S^4)$ is surjective as well, and the latter group is in the stable range. We show in Proposition 4L.11 that the stable group π_2^S is nonzero, and so we conclude that $\pi_{n+2}(S^n) \approx \mathbb{Z}_2$ for all $n \geq 2$, generated by the composition $(\Sigma^{n-2}\eta)(\Sigma^{n-1}\eta)$.

We will see in [SSAT] that the EHP sequence extends all the way to the left to form an infinite exact sequence when n is odd, and when n is even a weaker statement holds: The sequence extends after factoring out all odd torsion.

Replacing S^n by any $(n-1)$ -connected CW complex X , our derivation of the finite EHP sequence generalizes immediately to give an exact sequence

$$\pi_{3n-2}(X) \xrightarrow{\Sigma} \pi_{3n-1}(\Sigma X) \rightarrow \pi_{3n-2}(X \wedge X) \rightarrow \pi_{3n-3}(X) \xrightarrow{\Sigma} \pi_{3n-2}(\Sigma X) \rightarrow \cdots$$

using the fact that $J_2(X)/X = X \wedge X$.

The generalization of the results of this section to $\Omega^n \Sigma^n X$ turns out to be of some importance in homotopy theory. In case we do not get to this topic in [SSAT], the reader can begin to learn about it by looking at [Carlsson & Milgram 1995].

Exercise

1. Show that $\Omega \Sigma X$ for a nonconnected CW complex X reduces to the connected case by showing that each path-component of $\Omega \Sigma X$ is homotopy equivalent to $\Omega \Sigma(\bigvee_{\alpha} X_{\alpha})$ where the X_{α} 's are the components of X .

4.K The Dold–Thom Theorem

In the preceding section we studied the free monoid $J(X)$ generated by a space X , and in this section we take up its commutative analog, the free abelian monoid generated by X . This is the infinite symmetric product $SP(X)$ introduced briefly in §3.C. The main result will be a theorem of [Dold & Thom 1958] asserting that $\pi_* SP(X) \approx \tilde{H}_*(X; \mathbb{Z})$ for all connected CW complexes X . In particular this yields the surprising fact that $SP(S^n)$ is a $K(\mathbb{Z}, n)$, and more generally that the functor SP takes Moore spaces $M(G, n)$ to Eilenberg–MacLane spaces $K(G, n)$. This leads to the general result that for all connected CW complexes X , $SP(X)$ has the homotopy type of a product of Eilenberg–MacLane spaces. In other words, the k -invariants of $SP(X)$ are all trivial.

The main step in the proof of the Dold-Thom theorem will be to show that the homotopy groups $\pi_* SP(X)$ define a homology theory. An easy computation of the coefficient groups $\pi_* SP(S^n)$ will then show that this must be ordinary homology with $\mathbb{Z}$ coefficients. A new idea needed for the proof of the main step is the notion of a *quasifibration*, generalizing fibrations and fiber bundles. In order to establish a few basic facts about quasifibrations we first make a small detour to prove an essentially elementary fact about relative homotopy groups.

A Mayer-Vietoris Property of Homotopy Groups

In this subsection we will be concerned largely with relative homotopy groups, and it will be impossible to avoid the awkward fact that there is no really good way to define the relative π_0 . What we will do as a compromise is to take $\pi_0(X, A, x_0)$ to be the quotient set $\pi_0(X, x_0)/\pi_0(A, x_0)$. This at least allows the long exact sequence of homotopy groups for (X, A) to end with the terms

$$\pi_0(A, x_0) \rightarrow \pi_0(X, x_0) \rightarrow \pi_0(X, A, x_0) \rightarrow 0$$

An exercise for §4.1 shows that the five-lemma can be applied to the map of long exact sequences induced by a map $(X, A) \rightarrow (Y, B)$, provided the basepoint is allowed to vary. However, the long exact sequence of a triple cannot be extended through the π_0 terms with this definition, so one must proceed with some caution.

The excision theorem for homology involves a space X with subspaces A and B such that X is the union of the interiors of A and B . In this situation we call $(X; A, B)$ an **excisive triad**. By a map $f: (X; A, B) \rightarrow (Y; C, D)$ we mean $f: X \rightarrow Y$ with $f(A) \subset C$ and $f(B) \subset D$.

Proposition 4K.1. *Let $f: (X; A, B) \rightarrow (Y; C, D)$ be a map of excisive triads. If the induced maps $\pi_i(A, A \cap B) \rightarrow \pi_i(C, C \cap D)$ and $\pi_i(B, A \cap B) \rightarrow \pi_i(D, C \cap D)$ are bijections for $i < n$ and surjections for $i = n$, for all choices of basepoints, then the same is true of the induced maps $\pi_i(X, A) \rightarrow \pi_i(Y, C)$. By symmetry the conclusion holds also for the maps $\pi_i(X, B) \rightarrow \pi_i(Y, D)$.*

The corresponding statement for homology is a trivial consequence of excision which says that $H_i(X, A) \approx H_i(B, A \cap B)$ and $H_i(Y, C) \approx H_i(D, C \cap D)$, so it is not necessary to assume anything about the map $H_i(A, A \cap B) \rightarrow H_i(C, C \cap D)$. With the failure of excision for homotopy groups, however, it is not surprising that the assumption on $\pi_i(A, A \cap B) \rightarrow \pi_i(C, C \cap D)$ cannot be dropped. An example is provided by the quotient map $f: D^2 \rightarrow S^2$ collapsing ∂D^2 to the north pole of S^2 , with C and D the northern and southern hemispheres of S^2 , and A and B their preimages under f .

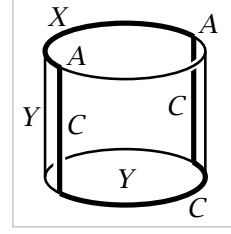
Proof: First we will establish a general fact about relative homotopy groups. Consider an inclusion $(X, A) \hookrightarrow (Y, C)$. We will show the following three conditions are equivalent for each $n \geq 1$:

(i) For all choices of basepoints the map $\pi_i(X, A) \rightarrow \pi_i(Y, C)$ induced by inclusion is surjective for $i = n$ and has trivial kernel for $i = n - 1$.

(ii) Let ∂D^n be written as the union of hemispheres $\partial_+ D^n$ and $\partial_- D^n$ intersecting in S^{n-2} . Then every map

$$(D^n \times \{0\} \cup \partial_+ D^n \times I, \partial_- D^n \times \{0\} \cup S^{n-2} \times I) \rightarrow (Y, C)$$

taking $(\partial_+ D^n \times \{1\}, S^{n-2} \times \{1\})$ to (X, A) extends to a map $(D^n \times I, \partial_- D^n \times I) \rightarrow (Y, C)$ taking $(D^n \times \{1\}, \partial_- D^n \times \{1\})$ to (X, A) .

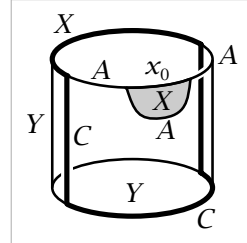


(iii) Condition (ii) with the added hypothesis that the restriction of the given map to $\partial_+ D^n \times I$ is independent of the I coordinate.

It is obvious that (ii) and (iii) are equivalent since the stronger hypothesis in (iii) can always be achieved by composing with a homotopy of $D^n \times I$ that shrinks $\partial_+ D^n \times I$ to $\partial_+ D^n \times \{1\}$.

To see that (iii) implies (i), let $f: (\partial_+ D^n \times \{1\}, S^{n-2} \times \{1\}) \rightarrow (X, A)$ represent an element of $\pi_{n-1}(X, A)$. If this is in the kernel of the map to $\pi_{n-1}(Y, C)$, then we get an extension of f over $D^n \times \{0\} \cup \partial_+ D^n \times I$, with the constant homotopy on $\partial_+ D^n \times I$ and $(D^n \times \{0\}, \partial_- D^n \times \{0\})$ mapping to (Y, C) . Condition (iii) then gives an extension over $D^n \times I$, whose restriction to $D^n \times \{1\}$ shows that f is zero in $\pi_{n-1}(X, A)$, so the kernel of $\pi_{n-1}(X, A) \rightarrow \pi_{n-1}(Y, C)$ is trivial. To check surjectivity of the map $\pi_n(X, A) \rightarrow \pi_n(Y, C)$, represent an element of $\pi_n(Y, C)$ by a map $f: D^n \times \{0\} \rightarrow Y$ taking $\partial_- D^n \times \{0\}$ to C and $\partial_+ D^n \times \{0\}$ to a chosen basepoint. Extend f over $\partial_+ D^n \times I$ via the constant homotopy, then extend over $D^n \times I$ by applying (iii). The result is a homotopy of the given f to a map representing an element of the image of $\pi_n(X, A) \rightarrow \pi_n(Y, C)$.

Now we show that (i) implies (ii). Given a map f as in the hypothesis of (ii), the injectivity part of (i) gives an extension of f over $D^n \times \{1\}$. Choose a small disk $E^n \subset \partial_- D^n \times I$, shown shaded in the figure, intersecting $\partial_- D^n \times \{1\}$ in a hemisphere $\partial_+ E^n$ of its boundary. We may assume the extended f has a constant value $x_0 \in A$ on $\partial_+ E^n$. Viewing the extended f as representing an element of $\pi_n(Y, C, x_0)$, the surjectivity part of (i) then gives an extension of f over $D^n \times I$ taking $(E^n, \partial_- E^n)$ to (X, A) and the rest of $\partial_- D^n \times I$ to C . The argument is finished by composing this extended f with a deformation of $D^n \times I$ pushing E^n into $D^n \times \{1\}$.



Having shown the equivalence of (i)–(iii), let us prove the proposition. We may reduce to the case that the given $f: (X; A, B) \rightarrow (Y; C, D)$ is an inclusion by using mapping cylinders. One's first guess would be to replace $(Y; C, D)$ by the triad of mapping cylinders $(M_f; M_{f|A}, M_{f|B})$, where we view $f|_A$ as a map $A \rightarrow C$ and $f|_B$ as a map $B \rightarrow D$. However, the triad $(M_f; M_{f|A}, M_{f|B})$ need not be excisive, for example if X

consists of two points A and B and Y is a single point. To remedy this problem, replace $M_{f|A}$ by its union with $f^{-1}(C) \times (1/2, 1)$ in M_f , and enlarge $M_{f|B}$ similarly.

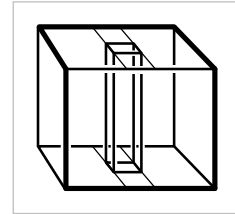
Now we prove the proposition for an inclusion $(X; A, B) \hookrightarrow (Y; C, D)$. The case $n = 0$ is trivial from the definitions, so let us assume $n \geq 1$. In view of the equivalence of condition (i) with (ii) and (iii), it suffices to show that condition (ii) for the inclusions $(A, A \cap B) \hookrightarrow (C, C \cap D)$ and $(B, A \cap B) \hookrightarrow (D, C \cap D)$ implies (iii) for the inclusion $(X, A) \hookrightarrow (Y, C)$. Let a map $f: D^n \times \{0\} \cup \partial_+ D^n \times I \rightarrow Y$ as in the hypothesis of (iii) be given. The argument will involve subdivision of D^n into smaller disks, and for this it is more convenient to use the cube I^n instead of D^n , so let us identify I^n with D^n in such a way that $\partial_- D^n$ corresponds to the face $I^{n-1} \times \{1\}$, which we denote by $\partial_- I^n$, and $\partial_+ D^n$ corresponds to the remaining faces of I^n , which we denote by $\partial_+ I^n$. Thus we are given f on $I^n \times \{0\}$ taking $\partial_+ I^n \times \{0\}$ to X and $\partial_- I^n \times \{0\}$ to C , and on $\partial_+ I^n \times I$ we have the constant homotopy.

Since $(Y; C, D)$ is an excisive triad, we can subdivide each of the I factors of $I^n \times \{0\}$ into subintervals so that f takes each of the resulting n -dimensional subcubes of $I^n \times \{0\}$ into either C or D . The extension of f we construct will have the following key property:

If K is a one of the subcubes of $I^n \times \{0\}$, or a lower-dimensional face of such a cube, then the extension of f takes $(K \times I, K \times \{1\})$ to (C, A) or (D, B) whenever f takes K to C or D , respectively.

Initially we have f defined on $\partial_+ I^n \times I$ with image in X , independent of the I coordinate, and we may assume the condition $(*)$ holds here since we may assume that $A = X \cap C$ and $B = X \cap D$, these conditions holding for the mapping cylinder construction described above.

Consider the problem of extending f over $K \times I$ for K one of the subcubes. We may assume that f has already been extended to $\partial_+ K \times I$ so that $(*)$ is satisfied, by induction on n and on the sequence of subintervals of the last coordinate of $I^n \times \{0\}$. To extend f over $K \times I$, let us first deal with the cases that the given f takes $(K, \partial_- K)$ to $(C, C \cap D)$ or $(D, C \cap D)$. Then by (ii) for the inclusion $(A, A \cap B) \hookrightarrow (C, C \cap D)$ or $(B, A \cap B) \hookrightarrow (D, C \cap D)$ we may extend f over $K \times I$ so that $(*)$ is still satisfied. If neither of these two cases applies, then the given f takes $(K, \partial_- K)$ just to (C, C) or (D, D) , and we can apply (ii) trivially to construct the desired extension of f over $K \times I$. $\square$



Corollary 4K.2. *Given a map $f: X \rightarrow Y$ and open covers $\{U_i\}$ of X and $\{V_i\}$ of Y with $f(U_i) \subset V_i$ for all i , then if each restriction $f: U_i \rightarrow V_i$ and more generally each $f: U_{i_1} \cap \cdots \cap U_{i_n} \rightarrow V_{i_1} \cap \cdots \cap V_{i_n}$ is a weak homotopy equivalence, so is $f: X \rightarrow Y$.*

Proof: First let us do the case of covers by two sets. By the five-lemma, the hypotheses imply that $\pi_n(U_i, U_1 \cap U_2) \rightarrow \pi_n(V_i, V_1 \cap V_2)$ is bijective for $i = 1, 2$, $n \geq 0$,

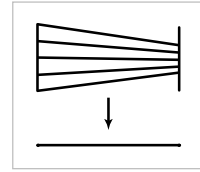
and all choices of basepoints. The preceding proposition then implies that the maps $\pi_n(X, U_1) \rightarrow \pi_n(Y, V_1)$ are isomorphisms. Hence by the five-lemma again, so are the maps $\pi_n(X) \rightarrow \pi_n(Y)$.

By induction, the case of finite covers by $k > 2$ sets reduces to the case of covers by two sets, by letting one of the two sets be the union of the first $k - 1$ of the given sets and the other be the k^{th} set. The case of infinite covers reduces to the finite case since for surjectivity of $\pi_n(X) \rightarrow \pi_n(Y)$, a map $S^n \rightarrow Y$ has compact image covered by finitely many V_i 's, and similarly for injectivity. $\square$

Quasifibrations

A map $p: E \rightarrow B$ with B path-connected is a **quasifibration** if the induced map $p_*: \pi_i(E, p^{-1}(b), x_0) \rightarrow \pi_i(B, b)$ is an isomorphism for all $b \in B$, $x_0 \in p^{-1}(b)$, and $i \geq 0$. We have shown in Theorem 4.41 that fiber bundles and fibrations have this property for $i > 0$, as a consequence of the homotopy lifting property, and the same reasoning applies for $i = 0$ since we assume B is path-connected.

For example, consider the natural projection $M_f \rightarrow I$ of the mapping cylinder of a map $f: X \rightarrow Y$. This projection will be a quasifibration iff f is a weak homotopy equivalence, since the latter condition is equivalent to having $\pi_i(M_f, p^{-1}(b)) = 0 = \pi_i(I, b)$ for all i and all $b \in I$. Note that if f is not surjective, there are paths in I that do not lift to paths in M_f with a prescribed starting point, so p will not be a fibration in such cases.



An alternative condition for a map $p: E \rightarrow B$ to be a quasifibration is that the inclusion of each fiber $p^{-1}(b)$ into the homotopy fiber F_b of p over b is a weak homotopy equivalence. Recall that F_b is the space of all pairs (x, γ) with $x \in E$ and γ a path in B from $p(x)$ to b . The actual fiber $p^{-1}(b)$ is included in F_b as the pairs with $x \in p^{-1}(b)$ and γ the constant path at x . To see the equivalence of the two definitions, consider the commutative triangle at the right, where $F_b \rightarrow E_p \rightarrow B$ is the usual path-fibration construction applied to p . The right-hand map in the diagram is an isomorphism for all $i \geq 0$, and the upper map will be an isomorphism for all $i \geq 0$ iff the inclusion $p^{-1}(b) \hookrightarrow F_b$ is a weak equivalence since $E \simeq E_p$. Hence the two definitions are equivalent.

$$\begin{array}{ccc} \pi_i(E, p^{-1}(b)) & \longrightarrow & \pi_i(E_p, F_b) \\ & \searrow & \swarrow \\ & \pi_i(B, b) & \end{array}$$

Recall from Proposition 4.61 that all fibers of a fibration over a path-connected base are homotopy equivalent. Since we are only considering quasifibrations over path-connected base spaces, this implies that all the fibers of a quasifibration have the same weak homotopy type. Quasifibrations over a base that is not path-connected are considered in the exercises, but we will not need this generality in what follows.

The following technical lemma gives various conditions for recognizing that a map is a quasifibration, which will be needed in the proof of the Dold–Thom theorem.

Lemma 4K.3. *A map $p : E \rightarrow B$ is a quasifibration if any one of the following conditions is satisfied:*

- (a) *B can be decomposed as the union of open sets V_1 and V_2 such that each of the restrictions $p^{-1}(V_1) \rightarrow V_1$, $p^{-1}(V_2) \rightarrow V_2$, and $p^{-1}(V_1 \cap V_2) \rightarrow V_1 \cap V_2$ is a quasifibration.*
- (b) *B is the union of an increasing sequence of subspaces $B_1 \subset B_2 \subset \cdots$, with the weak or direct limit topology, such that each restriction $p^{-1}(B_n) \rightarrow B_n$ is a quasifibration.*
- (c) *There is a deformation F_t of E into a subspace E' , covering a deformation $\bar{F}_t$ of B into a subspace B' , such that the restriction $E' \rightarrow B'$ is a quasifibration and $F_1 : p^{-1}(b) \rightarrow p^{-1}(\bar{F}_1(b))$ is a weak homotopy equivalence for each $b \in B$.*

By a ‘deformation’ in (c) we mean a deformation retraction in the weak sense as defined in the exercises for Chapter 0, where the homotopy is not required to be the identity on the subspace.

Proof: (a) To avoid some tedious details we will consider only the case that the fibers of p are path-connected, which will suffice for our present purposes, leaving the general case as an exercise for the reader. This hypothesis on fibers guarantees that all π_0 ’s arising in the proof are trivial. In particular, by an exercise for §4.1 this allows us to terminate long exact sequences of homotopy groups of triples with zeros in the π_0 positions.

Let $U_1 = p^{-1}(V_1)$ and $U_2 = p^{-1}(V_2)$. By applying the five-lemma to the long exact sequences of homotopy groups of the triples $(U_k, U_1 \cap U_2, p^{-1}(b))$ and $(V_k, V_1 \cap V_2, b)$ we deduce that the maps $\pi_i(U_k, U_1 \cap U_2) \rightarrow \pi_i(V_k, V_1 \cap V_2)$ are isomorphisms for $k = 1, 2$ and all i . Then Proposition 4K.1 implies that the maps $\pi_i(E, U_1) \rightarrow \pi_i(B, V_1)$ are isomorphisms. The hypothesis that the maps $\pi_i(U_1, p^{-1}(b)) \rightarrow \pi_i(V_1, b)$ are isomorphisms then implies via the five-lemma that the maps $\pi_i(E, p^{-1}(b)) \rightarrow \pi_i(B, b)$ are isomorphisms.

(b) This is the standard compactness argument for direct limits. Every compact subspace of E lies in one of the subspaces $E_n = p^{-1}(B_n)$, so $\pi_i(E, p^{-1}(b))$ is the direct limit $\varinjlim \pi_i(E_n, p^{-1}(b))$, hence the map $\pi_i(E, p^{-1}(b)) \rightarrow \pi_i(B, b)$ is an isomorphism if each map $\pi_i(E_n, p^{-1}(b)) \rightarrow \pi_i(B, b)$ is an isomorphism.

(c) Consider the commutative diagram

$$\begin{array}{ccccc} (E, p^{-1}(b)) & \xrightarrow{F_1} & (E', p^{-1}(\bar{F}_1(b))) & \longrightarrow & (E, p^{-1}(\bar{F}_1(b))) \\ \downarrow & & \downarrow & & \\ (B, b) & \xrightarrow{\bar{F}_1} & (B', \bar{F}_1(b)) & & \end{array}$$

where the upper right map is inclusion. The composition across the top of the diagram induces an isomorphism on relative homotopy groups via the five-lemma and the hypothesis that $F_1 : p^{-1}(b) \rightarrow p^{-1}(\bar{F}_1(b))$ is a weak homotopy equivalence. Since the inclusion $E' \hookrightarrow E$ is a homotopy equivalence, the upper right map also induces

isomorphisms on relative homotopy groups. Hence the upper left map F_1 induces isomorphisms on relative homotopy groups. The lower map $\bar{F}_1$ is a homotopy equivalence, so from commutativity of the diagram we deduce that the left-hand vertical map induces isomorphisms on homotopy groups since this is true of the right-hand vertical map by assumption. $\square$

Symmetric Products

Let us recall the definition from §3.C. For a space X the n -fold symmetric product $SP_n(X)$ is the quotient space of the product of n copies of X obtained by factoring out the action of the symmetric group permuting the factors. A choice of basepoint $e \in X$ gives inclusions $SP_n(X) \hookrightarrow SP_{n+1}(X)$ induced by $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, e)$, and $SP(X)$ is defined to be the union of this increasing sequence of spaces, with the direct limit topology. Note that SP_n is a homotopy functor: A map $f: X \rightarrow Y$ induces $f_*: SP_n(X) \rightarrow SP_n(Y)$, and $f \simeq g$ implies $f_* \simeq g_*$. Hence $X \simeq Y$ implies $SP_n(X) \simeq SP_n(Y)$. In similar fashion SP is a homotopy functor on the category of basepointed spaces and basepoint-preserving homotopy classes of maps. It follows that $X \simeq Y$ implies $SP(X) \simeq SP(Y)$ for connected CW complexes X and Y since in this case requiring maps and homotopies to preserve basepoints does not affect the relation of homotopy equivalence.

Example 4K.4. An interesting special case is when $X = S^2$ because in this case $SP(S^2)$ can be identified with $\mathbb{CP}^\infty$ in the following way. We first identify $\mathbb{CP}^n$ with the nonzero polynomials of degree at most n with coefficients in $\mathbb{C}$, modulo scalar multiplication, by letting $a_0 + \dots + a_n z^n$ correspond to the line containing $(a_0, \dots, a_n)$. The sphere S^2 we view as $\mathbb{C} \cup \{\infty\}$, and then we define $f: (S^2)^n \rightarrow \mathbb{CP}^n$ by setting $f(a_1, \dots, a_n) = (z + a_1) \cdots (z + a_n)$ with factors $z + \infty$ omitted, so in particular $f(\infty, \dots, \infty) = 1$. To check that f is continuous, suppose some a_i approaches ∞ , say a_n , and all the other a_j 's are finite. Then if we write

$$(z + a_1) \cdots (z + a_n) = z^n + (a_1 + \dots + a_n)z^{n-1} + \dots + \sum_{i_1 < \dots < i_k} a_{i_1} \cdots a_{i_k} z^{n-k} + \dots + a_1 \cdots a_n$$

we see that, dividing through by a_n and letting a_n approach ∞ , this polynomial approaches $z^{n-1} + (a_1 + \dots + a_{n-1})z^{n-2} + \dots + a_1 \cdots a_{n-1} = (z + a_1) \cdots (z + a_{n-1})$. The same argument would apply if several a_i 's approach ∞ simultaneously.

The value $f(a_1, \dots, a_n)$ is unchanged under permutation of the a_i 's, so there is an induced map $SP_n(S^2) \rightarrow \mathbb{CP}^n$ which is a continuous bijection, hence a homeomorphism since both spaces are compact Hausdorff. Letting n go to ∞ , we then get a homeomorphism $SP(S^2) \approx \mathbb{CP}^\infty$.

The same argument can be used to show that $SP_n(S^1) \simeq S^1$ for all n , including $n = \infty$. Namely, the argument shows that $SP_n(\mathbb{C} - \{0\})$ can be identified with the

polynomials $z^n + a_{n-1}z^{n-1} + \cdots + a_0$ with $a_0 \neq 0$, or in other words, the n -tuples $(a_0, \dots, a_{n-1}) \in \mathbb{C}^n$ with $a_0 \neq 0$, and this subspace of $\mathbb{C}^n$ deformation retracts onto a circle.

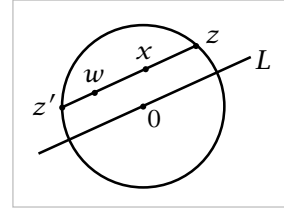
The symmetric products of higher-dimensional spheres are more complicated, though things are not so bad for the 2-fold symmetric product:

Example 4K.5. Let us show that $SP_2(S^n)$ is homeomorphic to the mapping cone of a map $\Sigma^n \mathbb{R}P^{n-1} \rightarrow S^n$. Hence $H_*(SP_2(S^n)) \approx H_*(S^n) \oplus \tilde{H}_*(\Sigma^{n+1} \mathbb{R}P^{n-1})$ from the long exact sequence of homology groups for the pair $(SP_2(S^n), S^n)$, since $SP_2(S^n)/S^n$ is $\Sigma^{n+1} \mathbb{R}P^{n-1}$ with no reduced homology below dimension $n+2$.

If we view S^n as $D^n/\partial D^n$, then $SP_2(S^n)$ becomes a certain quotient of $D^n \times D^n$. Viewing $D^n \times D^n$ as the cone on its boundary $D^n \times \partial D^n \cup \partial D^n \times D^n$, the identifications that produce $SP_2(S^n)$ respect the various concentric copies of this boundary which fill up the interior of $D^n \times D^n$, so it suffices to analyze the identifications in all these copies of the boundary. The identifications on the boundary of $D^n \times D^n$ itself yield S^n . This is clear since the identification $(x, y) \sim (y, x)$ converts $D^n \times \partial D^n \cup \partial D^n \times D^n$ to $D^n \times \partial D^n$, and all points of ∂D^n are identified in S^n .

It remains to see that the identifications $(x, y) \sim (y, x)$ on each concentric copy of the boundary in the interior of $D^n \times D^n$ produce $\Sigma^n \mathbb{R}P^{n-1}$. Denote by Z the quotient of $D^n \times \partial D^n \cup \partial D^n \times D^n$ under these identifications. This is the same as the quotient of $D^n \times \partial D^n$ under the identifications $(x, y) \sim (y, x)$ for $(x, y) \in \partial D^n \times \partial D^n$.

Define $f: D^n \times \mathbb{R}P^{n-1} \rightarrow Z$ by $f(x, L) = (w, z)$ where w and z are defined as in the figure, that is, x is equidistant from $z \in \partial D^n$ and $w \in D^n$ along the line through x parallel to L . Note that if x happens to be the midpoint of the segment zz' , then $w = z'$ and there is no way to distinguish between w and z , but since f takes values in the quotient space Z , this is not a problem. If $x \in \partial D^n$ then $w = z = x$, independent of L . If $x \in D^n - \partial D^n$ then $w \neq z$, and conversely, given $(w, z) \in D^n \times \partial D^n$ with $w \neq z$ there is a unique (x, L) with $f(x, L) = (w, z)$, namely x is the midpoint of the segment wz and L is the line parallel to this segment. In view of these remarks, we see that Z is the quotient space of $D^n \times \mathbb{R}P^{n-1}$ under the identifications $(x, L) \sim (x, L')$ if $x \in \partial D^n$. This quotient is precisely $\Sigma^n \mathbb{R}P^{n-1}$.



This example illustrates that passing from a CW structure on X to a CW structure on $SP_n(X)$ or $SP(X)$ is not at all straightforward. However, if X is a simplicial complex, there is a natural way of putting Δ -complex structures on $SP_n(X)$ and $SP(X)$, as follows. A simplicial complex structure on X gives a CW structure on the product of n copies of X , with cells n -fold products of simplices. Such a product has a canonical barycentric subdivision as a simplicial complex, with vertices the points whose coordinates are barycenters of simplices of X . By induction over skeleta, this